ASSIGNMENT NO.1]

SET-A]

1. **Create a Simple Application Which Shows Life Cycle of Activity.**

->

MainActivity.java

```
package com.example.androidsolution;

import android.os.Bundle;
import android.util.Log;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    String TAG = "ActivityLifecycle";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        showToast("onCreate()");
        Log.d(TAG, "onCreate()");
    }

    @Override
    protected void onStart() {
        super.onStart();
        showToast("onStart()");
        Log.d(TAG, "onStart()");
    }

    @Override
    protected void onResume() {
        super.onResume();
        showToast("onResume()");
        Log.d(TAG, "onResume()");
    }

    @Override
```

```
    protected void onPause() {
        super.onPause();
        showToast("onPause()");
        Log.d(TAG, "onPause()");
    }

    @Override
    protected void onStop() {
        super.onStop();
        showToast("onStop()");
        Log.d(TAG, "onStop()");
    }

    @Override
    protected void onRestart() {
        super.onRestart();
        showToast("onRestart()");
        Log.d(TAG, "onRestart()");
    }

    @Override
    protected void onDestroy() {
        super.onDestroy();
        showToast("onDestroy()");
        Log.d(TAG, "onDestroy()");
    }

    private void showToast(String message) {
        Toast.makeText(this, message, Toast.LENGTH_SHORT).show();
    }
}
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:id="@+id/lifecycle_text"
        android:layout_width="wrap_content"
```

```
    android:layout_height="wrap_content"
    android:text="Activity Lifecycle Demo"
    android:textSize="20sp"
    android:textStyle="bold"
    android:textColor="#000000" />
</LinearLayout>
```

---

2. Create a Simple Application Which Send -Hellol message from one activity to another with help of Button (Use Intent).

->

MainActivity.java

```
package com.example.hello;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

  Button sendButton;

  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    sendButton = findViewById(R.id.btnSend);

    sendButton.setOnClickListener(new View.OnClickListener() {
      @Override
      public void onClick(View v) {
        Intent intent = new Intent(MainActivity.this, SecondActivity.class);
        intent.putExtra("message", "Hello!");
        startActivity(intent);
      }
```

```
        });
    }
}
```

SecondActivity.java

```
package com.example.hello;


import android.os.Bundle;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;

public class SecondActivity extends AppCompatActivity {

    TextView textReceived;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_second);

        textReceived = findViewById(R.id.txtMessage);

        // Get the message from Intent
        String message = getIntent().getStringExtra("message");

        // Show the message
        textReceived.setText(message);
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="16dp">
```

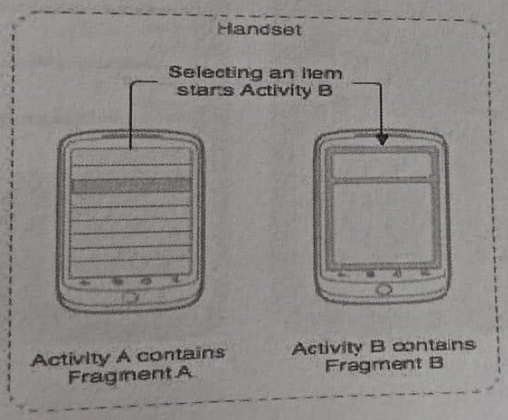
```
    <Button
        android:id="@+id/btnSend"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Send Hello!" />
</LinearLayout>
```

activity_second.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="16dp">

    <TextView
        android:id="@+id/txtMessage"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="20sp"
        android:textStyle="bold" />
</LinearLayout>
```

---

3. Create Simple application to display details of selected list item on Second Activity (use Fragmentation).



->

DetailActivity.java

```
package com.example.fragmentex;

import android.os.Bundle;

import androidx.appcompat.app.AppCompatActivity;

public class DetailActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_detail);

        String detail = getIntent().getStringExtra("item_detail");

        FragmentB fragmentB = new FragmentB();

        // Pass data to FragmentB
        Bundle bundle = new Bundle();
        bundle.putString("detail_key", detail);
        fragmentB.setArguments(bundle);

        getSupportFragmentManager()
                .beginTransaction()
                .replace(R.id.detail_fragment_container, fragmentB)
                .commit();
```

```
    }
}
```

FragmentA.java

```
package com.example.fragmentex;

import android.content.Intent;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.ArrayAdapter;
import android.widget.ListView;

import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;

public class FragmentA extends Fragment {

    // Optional parameters (if needed)
    private static final String ARG_PARAM1 = "param1";
    private static final String ARG_PARAM2 = "param2";
    private String mParam1;
    private String mParam2;

    String[] items = {"Item 1", "Item 2", "Item 3"};

    public FragmentA() {
        // Required empty public constructor
    }

    public static FragmentA newInstance(String param1, String param2) {
        FragmentA fragment = new FragmentA();
        Bundle args = new Bundle();
        args.putString(ARG_PARAM1, param1);
        args.putString(ARG_PARAM2, param2);
        fragment.setArguments(args);
        return fragment;
    }

    @Override
```

```
    public void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        if (getArguments() != null) {
            mParam1 = getArguments().getString(ARG_PARAM1);
            mParam2 = getArguments().getString(ARG_PARAM2);
        }
    }

    @Nullable
    @Override
    public View onCreateView(@NonNull LayoutInflater inflater,
                             @Nullable ViewGroup container,
                             @Nullable Bundle savedInstanceState) {

        View view = inflater.inflate(R.layout.fragment_a, container, false);

        ListView listView = view.findViewById(R.id.list_view);
        ArrayAdapter<String> adapter = new ArrayAdapter<>(
                getActivity(), android.R.layout.simple_list_item_1, items);
        listView.setAdapter(adapter);

        listView.setOnItemClickListener((parent, view1, position, id) -> {
            Intent intent = new Intent(getActivity(), DetailActivity.class);
            intent.putExtra("item_detail", items[position]);
            startActivity(intent);
        });

        return view;
    }
}
```

FragmentB.java

```
package com.example.fragmentex;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;

import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
```

```
import androidx.fragment.app.Fragment;

public class FragmentB extends Fragment {
   @Nullable
   @Override
   public View onCreateView(@NonNull LayoutInflater inflater,
                            @Nullable ViewGroup container,
                            @Nullable Bundle savedInstanceState) {
      View view = inflater.inflate(R.layout.fragment_b, container, false);

      TextView textView = view.findViewById(R.id.detail_text);
      String detail = getArguments().getString("detail_key", "No detail");
      textView.setText(detail);

      return view;
   }
}
```

MainActivity.java

```
package com.example.fragmentex;

import android.os.Bundle;

import androidx.appcompat.app.AppCompatActivity;
import androidx.fragment.app.FragmentManager;

public class MainActivity extends AppCompatActivity {
   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);

      // Load Fragment A
      getSupportFragmentManager()
            .beginTransaction()
            .replace(R.id.fragment_container, new FragmentA())
            .commit();
   }
}
```

activity_detail.xml

```
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:id="@+id/detail_fragment_container"
   android:layout_width="match_parent"
   android:layout_height="match_parent" />
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:id="@+id/fragment_container"
   android:layout_width="match_parent"
   android:layout_height="match_parent"/>
```

fragment_a.xml

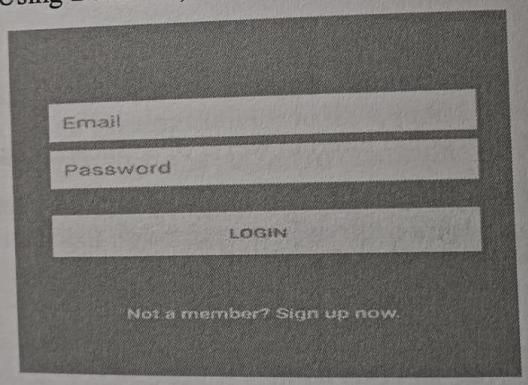
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:orientation="vertical"
   android:layout_width="match_parent"
   android:layout_height="match_parent">

   <ListView
      android:id="@+id/list_view"
      android:layout_width="match_parent"
      android:layout_height="match_parent" />
</LinearLayout>
```

fragment_b.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:orientation="vertical"
   android:padding="16dp"
   android:layout_width="match_parent"
   android:layout_height="match_parent">

   <TextView
      android:id="@+id/detail_text"
      android:textSize="20sp"
      android:layout_width="match_parent"
      android:layout_height="wrap_content" />
</LinearLayout>
```

SET B

1. Create simple application with Login Screen. On successful login, gives message go to next Activity (Without Using Database).



->

MainActivity.java

```
package com.example.loginscreen;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText emailInput, passwordInput;
    Button loginBtn;

    final String correctEmail = "user@example.com";
    final String correctPassword = "1234";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
```

```
        emailInput = findViewById(R.id.email);
        passwordInput = findViewById(R.id.password);
        loginBtn = findViewById(R.id.login_button);

        loginBtn.setOnClickListener(v -> {
            String email = emailInput.getText().toString();
            String password = passwordInput.getText().toString();

            if (email.equals(correctEmail) && password.equals(correctPassword)) {
                Toast.makeText(MainActivity.this, "Login Successful!",
Toast.LENGTH_SHORT).show();
                startActivity(new Intent(MainActivity.this, WelcomeActivity.class));
            } else {
                Toast.makeText(MainActivity.this, "Invalid Email or Password",
Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

WelcomeActivity.java

```
package com.example.loginscreen;


import android.os.Bundle;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;

public class WelcomeActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_welcome);

        TextView text = findViewById(R.id.welcome_text);
        text.setText("Welcome to the next screen!");
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp">

    <EditText
        android:id="@+id/email"
        android:hint="Email"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:inputType="textEmailAddress"/>

    <EditText
        android:id="@+id/password"
        android:hint="Password"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:inputType="textPassword"
        android:layout_marginTop="16dp"/>

    <Button
        android:id="@+id/login_button"
        android:text="LOGIN"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="24dp"/>
</LinearLayout>
```

activity_welcome.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="24dp">

    <TextView
        android:id="@+id/welcome_text"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```

```
    android:text="Welcome!"
    android:textSize="24sp"/>
</LinearLayout>
```

2.Create First Activity to accept information like Student First Name, Middle Name, Last Name, Date of birth, Address, Email ID and display all information on Second Activity when user click on Submit button.

->

DisplayActivity.java

```
package com.example.studentinfo;


import android.os.Bundle;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;

public class DisplayActivity extends AppCompatActivity {
   TextView infoText;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_display);

      infoText = findViewById(R.id.info_text);

      String data = "First Name: " + getIntent().getStringExtra("first_name") + "\n"
            + "Middle Name: " + getIntent().getStringExtra("middle_name") + "\n"
            + "Last Name: " + getIntent().getStringExtra("last_name") + "\n"
            + "DOB: " + getIntent().getStringExtra("dob") + "\n"
            + "Address: " + getIntent().getStringExtra("address") + "\n"
            + "Email: " + getIntent().getStringExtra("email");

      infoText.setText(data);
   }
}
```

ManiActivity.java

```
package com.example.studentinfo;

import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText firstName, middleName, lastName, dob, address, email;
    Button submitBtn;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        firstName = findViewById(R.id.first_name);
        middleName = findViewById(R.id.middle_name);
        lastName = findViewById(R.id.last_name);
        dob = findViewById(R.id.dob);
        address = findViewById(R.id.address);
        email = findViewById(R.id.email);
        submitBtn = findViewById(R.id.submit_btn);

        submitBtn.setOnClickListener(v -> {
            Intent intent = new Intent(MainActivity.this, DisplayActivity.class);
            intent.putExtra("first_name", firstName.getText().toString());
            intent.putExtra("middle_name", middleName.getText().toString());
            intent.putExtra("last_name", lastName.getText().toString());
            intent.putExtra("dob", dob.getText().toString());
            intent.putExtra("address", address.getText().toString());
            intent.putExtra("email", email.getText().toString());
            startActivity(intent);
        });
    }
}
```

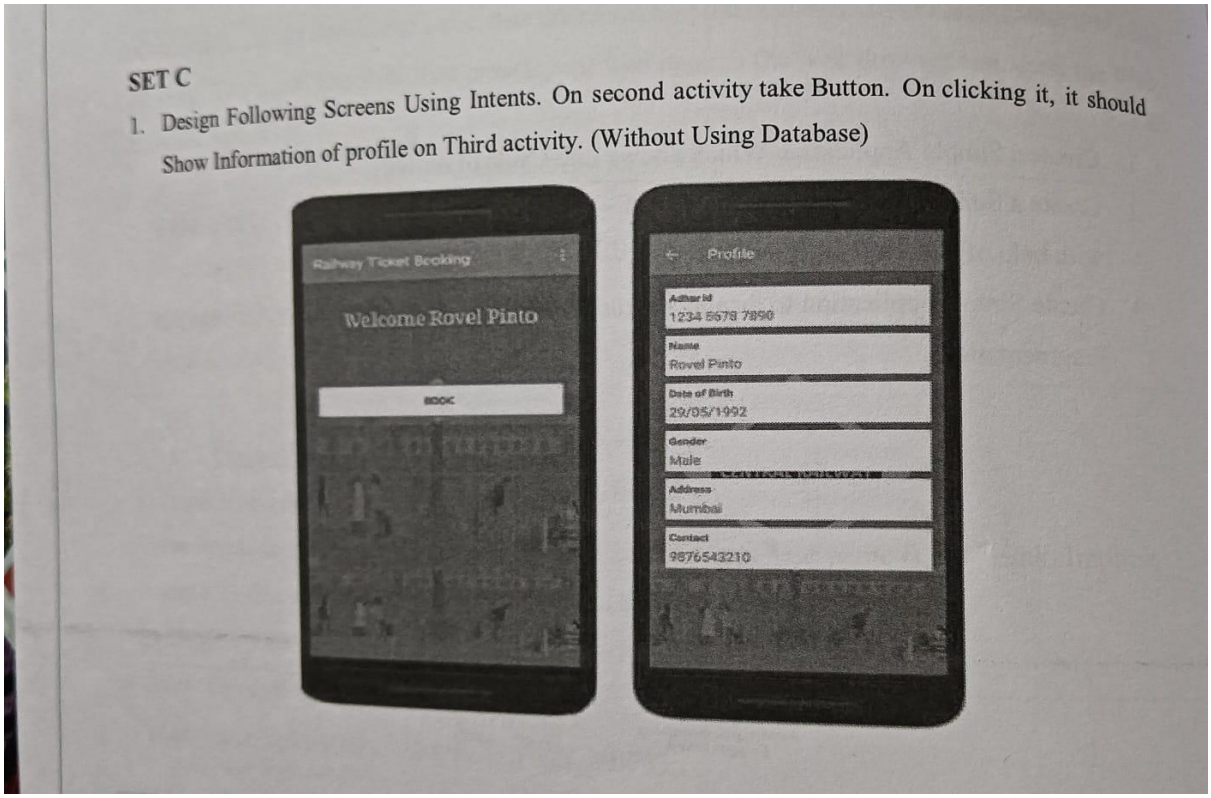
activity_display.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent">

   <TextView
      android:id="@+id/info_text"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:textSize="18sp"
      android:padding="24dp"/>
</ScrollView>
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent">

   <LinearLayout
      android:orientation="vertical"
      android:padding="24dp"
      android:layout_width="match_parent"
      android:layout_height="wrap_content">

      <EditText android:id="@+id/first_name" android:hint="First Name"
         android:layout_width="match_parent" android:layout_height="wrap_content"/>
      <EditText android:id="@+id/middle_name" android:hint="Middle Name"
         android:layout_width="match_parent" android:layout_height="wrap_content"/>
      <EditText android:id="@+id/last_name" android:hint="Last Name"
         android:layout_width="match_parent" android:layout_height="wrap_content"/>
      <EditText android:id="@+id/dob" android:hint="Date of Birth"
         android:layout_width="match_parent" android:layout_height="wrap_content"/>
      <EditText android:id="@+id/address" android:hint="Address"
         android:layout_width="match_parent" android:layout_height="wrap_content"/>
      <EditText android:id="@+id/email" android:hint="Email ID"
         android:inputType="textEmailAddress"
         android:layout_width="match_parent" android:layout_height="wrap_content"/>

      <Button android:id="@+id/submit_btn" android:text="Submit"
         android:layout_marginTop="20dp"
         android:layout_width="match_parent" android:layout_height="wrap_content"/>
```

```
    </LinearLayout>
</ScrollView>
```



SET C

1. Design Following Screens Using Intents. On second activity take Button. On clicking it, it should Show Information of profile on Third activity. (Without Using Database)

->

<u>MainActivity.java</u>

```
package com.example.intentexample;

import android.content.Intent;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        Intent intent = new Intent(this, SecondActivity.class);
        intent.putExtra("name", "Rovel Pinto");
        startActivity(intent);
        finish();
    }
}
```

SecondActivity.java

```
package com.example.intentexample;



import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class SecondActivity extends AppCompatActivity {
    TextView welcomeText;
    Button bookBtn;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_second);

        welcomeText = findViewById(R.id.welcomeText);
        bookBtn = findViewById(R.id.bookButton);

        String name = getIntent().getStringExtra("name");
        welcomeText.setText("Welcome " + name);

        bookBtn.setOnClickListener(v -> {
            Intent intent = new Intent(SecondActivity.this, ThirdActivity.class);
            intent.putExtra("aadhaar", "1234 5678 7890");
            intent.putExtra("name", "Rovel Pinto");
            intent.putExtra("dob", "29/05/1992");
            intent.putExtra("gender", "Male");
            intent.putExtra("address", "Mumbai");
            intent.putExtra("contact", "9876543210");
            startActivity(intent);
        });
    }
}
```

ThirdActivity.java

```
package com.example.intentexample;
```

```
import android.os.Bundle;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class ThirdActivity extends AppCompatActivity {
   TextView aadhaar, name, dob, gender, address, contact;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_third);

      aadhaar = findViewById(R.id.textAadhaar);
      name = findViewById(R.id.textName);
      dob = findViewById(R.id.textDob);
      gender = findViewById(R.id.textGender);
      address = findViewById(R.id.textAddress);
      contact = findViewById(R.id.textContact);

      aadhaar.setText("Aadhaar Id: " + getIntent().getStringExtra("aadhaar"));
      name.setText("Name: " + getIntent().getStringExtra("name"));
      dob.setText("Date of Birth: " + getIntent().getStringExtra("dob"));
      gender.setText("Gender: " + getIntent().getStringExtra("gender"));
      address.setText("Address: " + getIntent().getStringExtra("address"));
      contact.setText("Contact: " + getIntent().getStringExtra("contact"));
   }
}
```

activity_main.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent" android:layout_height="match_parent"
   android:orientation="vertical"/>
```

activity_second.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:orientation="vertical" android:layout_width="match_parent"
   android:layout_height="match_parent" android:padding="16dp">

   <TextView
      android:id="@+id/welcomeText"
```

```
        android:textSize="20sp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content" />

    <Button
        android:id="@+id/bookButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="BOOK" />
</LinearLayout>
```

activity_third.xml

```
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:orientation="vertical" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:padding="16dp">

        <TextView android:id="@+id/textAadhaar" android:layout_width="wrap_content"
android:layout_height="wrap_content"/>
        <TextView android:id="@+id/textName" android:layout_width="wrap_content"
android:layout_height="wrap_content"/>
        <TextView android:id="@+id/textDob" android:layout_width="wrap_content"
android:layout_height="wrap_content"/>
        <TextView android:id="@+id/textGender" android:layout_width="wrap_content"
android:layout_height="wrap_content"/>
        <TextView android:id="@+id/textAddress" android:layout_width="wrap_content"
android:layout_height="wrap_content"/>
        <TextView android:id="@+id/textContact" android:layout_width="wrap_content"
android:layout_height="wrap_content"/>
    </LinearLayout>
</ScrollView>
```

ASSIGNMENT NO.2

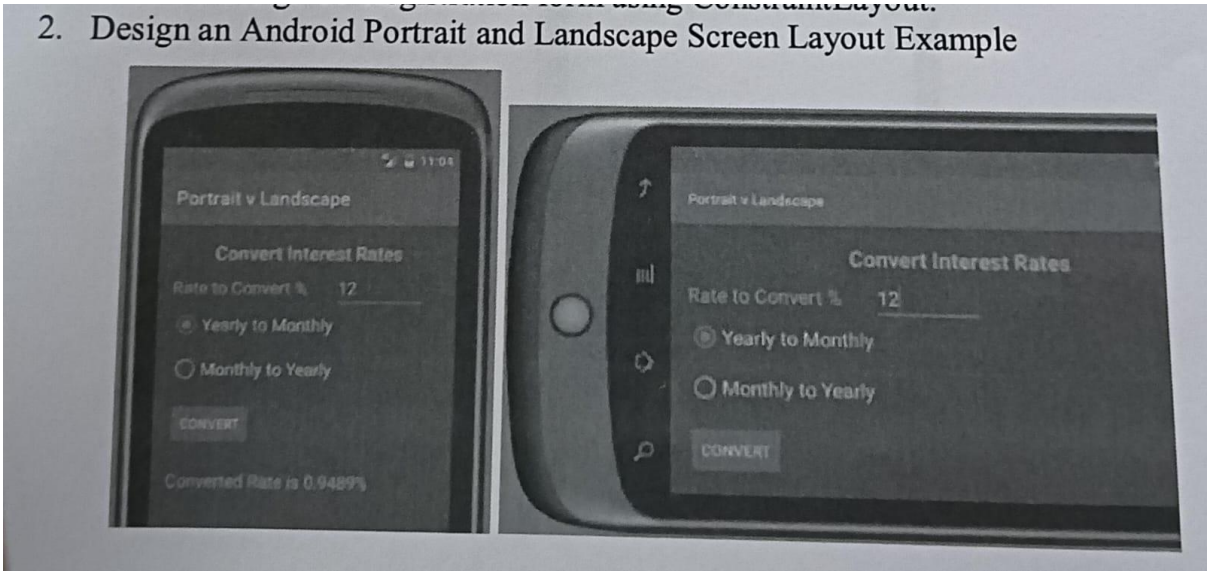
1. Create a design of Registration form using ConstraintLayout.

-> In this code only add **activity_registeration.xml** file .No more add files like MainActivity and activity_main.xml .

activity_registeration.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="24dp">

    <TextView
        android:id="@+id/tvTitle"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Register"
        android:textSize="24sp"
        android:textStyle="bold"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent" />

    <EditText
        android:id="@+id/etFullName"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:hint="Full Name"
        app:layout_constraintTop_toBottomOf="@id/tvTitle"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="24dp" />

    <EditText
        android:id="@+id/etEmail"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:hint="Email"
        android:inputType="textEmailAddress"
        app:layout_constraintTop_toBottomOf="@id/etFullName"
        app:layout_constraintStart_toStartOf="parent"
```

```
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="16dp" />

    <EditText
        android:id="@+id/etPassword"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:hint="Password"
        android:inputType="textPassword"
        app:layout_constraintTop_toBottomOf="@id/etEmail"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="16dp" />

    <EditText
        android:id="@+id/etConfirmPassword"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:hint="Confirm Password"
        android:inputType="textPassword"
        app:layout_constraintTop_toBottomOf="@id/etPassword"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="16dp" />

    <Button
        android:id="@+id/btnRegister"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:text="Register"
        app:layout_constraintTop_toBottomOf="@id/etConfirmPassword"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="24dp" />

</androidx.constraintlayout.widget.ConstraintLayout>
```



->

1] Step 1: Default layout (Portrait)

MainActivity.java

```
package com.example.convertinterest;

import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.view.View;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        View rootView = findViewById(android.R.id.content);
        ViewCompat.setOnApplyWindowInsetsListener(rootView, (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
```

```
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp">

    <TextView
        android:id="@+id/tvTitle"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Convert Interest Rates"
        android:textSize="20sp"
        android:textStyle="bold"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent" />

    <TextView
        android:id="@+id/tvRate"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Rate to Convert %"
        app:layout_constraintTop_toBottomOf="@id/tvTitle"
        app:layout_constraintStart_toStartOf="parent"
        android:layout_marginTop="24dp" />

    <EditText
        android:id="@+id/etRate"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:inputType="numberDecimal"
        android:text="12"
        app:layout_constraintStart_toEndOf="@id/tvRate"
        app:layout_constraintTop_toTopOf="@id/tvRate"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginStart="8dp" />

    <RadioGroup
```

```
        android:id="@+id/radioGroup"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        app:layout_constraintTop_toBottomOf="@id/etRate"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="16dp">

        <RadioButton
            android:id="@+id/rbYearToMonth"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Yearly to Monthly" />

        <RadioButton
            android:id="@+id/rbMonthToYear"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Monthly to Yearly" />
    </RadioGroup>

    <Button
        android:id="@+id/btnConvert"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="CONVERT"
        app:layout_constraintTop_toBottomOf="@id/radioGroup"
        app:layout_constraintStart_toStartOf="parent"
        android:layout_marginTop="16dp" />

    <TextView
        android:id="@+id/tvResult"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:text="Converted Rate is 0.9489%"
        app:layout_constraintTop_toBottomOf="@id/btnConvert"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="16dp" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

Step 2: Landscape layout

MainActivity.java

```
package com.example.convertinterest;

import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.view.View;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        View rootView = findViewById(android.R.id.content);
        ViewCompat.setOnApplyWindowInsetsListener(rootView, (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}
```

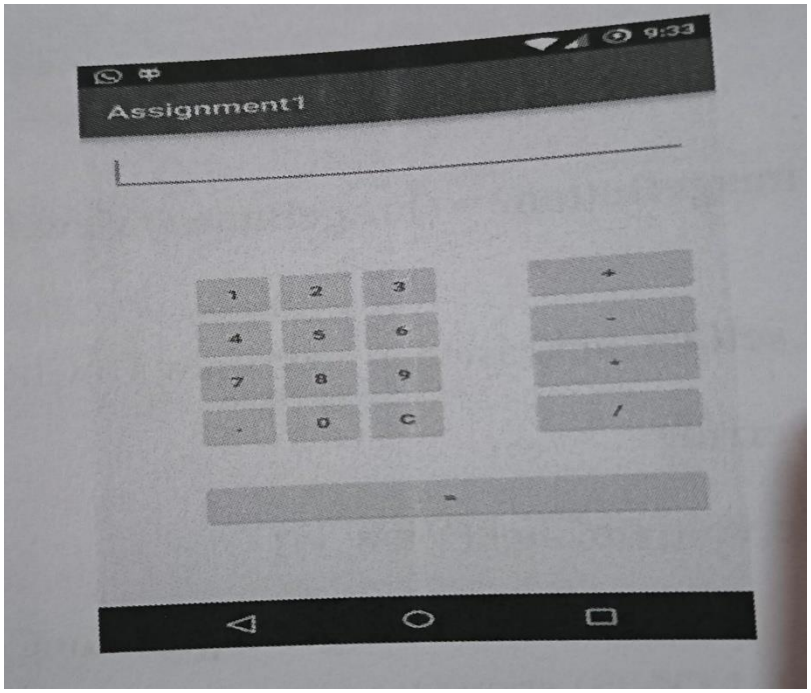
activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp">

    <TextView
        android:id="@+id/tvTitle"
```

```
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Convert Interest Rates"
    android:textSize="20sp"
    android:textStyle="bold"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintStart_toStartOf="parent" />

<TextView
    android:id="@+id/tvRate"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Rate to Convert %"
    app:layout_constraintTop_toBottomOf="@id/tvTitle"
    app:layout_constraintStart_toStartOf="parent"
    android:layout_marginTop="24dp" />

<EditText
    android:id="@+id/etRate"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:inputType="numberDecimal"
    android:text="12"
    app:layout_constraintStart_toEndOf="@id/tvRate"
    app:layout_constraintTop_toTopOf="@id/tvRate"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_marginStart="8dp" />

<RadioGroup
    android:id="@+id/radioGroup"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    app:layout_constraintTop_toBottomOf="@id/tvRate"
    app:layout_constraintStart_toStartOf="parent"
    android:layout_marginTop="24dp">

    <RadioButton
        android:id="@+id/rbYearToMonth"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Yearly to Monthly" />

    <RadioButton
```

```
        android:id="@+id/rbMonthToYear"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Monthly to Yearly" />
    </RadioGroup>

    <Button
      android:id="@+id/btnConvert"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="CONVERT"
      app:layout_constraintTop_toBottomOf="@id/radioGroup"
      app:layout_constraintStart_toStartOf="parent"
      android:layout_marginTop="16dp" />

    <TextView
      android:id="@+id/tvResult"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Converted Rate is 0.9489%"
      app:layout_constraintStart_toEndOf="@id/radioGroup"
      app:layout_constraintTop_toTopOf="@id/radioGroup"
      android:layout_marginStart="24dp" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

3] Create the simple calculator shown below also perform appropriate operation



->

<u>MainActivity.java</u>

```
package com.example.simplecalculator;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    EditText editText;
    String input = "";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        editText = findViewById(R.id.editText);
    }

    public void buttonClick(View view) {
        Button button = (Button) view;
```

```
        input += button.getText().toString();
        editText.setText(input);
    }

    public void clearClick(View view) {
        input = "";
        editText.setText("");
    }

    public void calculateClick(View view) {
        try {
            double result = evaluate(input);
            editText.setText(String.valueOf(result));
            input = String.valueOf(result);
        } catch (Exception e) {
            editText.setText("Error");
            input = "";
        }
    }

    // Expression evaluator: handles +, -, *, /, parentheses, and decimals
    private double evaluate(String expression) {
        return new Object() {
            int pos = -1, ch;

            void nextChar() {
                ch = (++pos < expression.length()) ? expression.charAt(pos) : -1;
            }

            boolean eat(int charToEat) {
                while (ch == ' ') nextChar();
                if (ch == charToEat) {
                    nextChar();
                    return true;
                }
                return false;
            }

            double parse() {
                nextChar();
                double x = parseExpression();
                if (pos < expression.length()) throw new RuntimeException("Unexpected: " + (char)
ch);
                return x;
```

```
        }

        double parseExpression() {
            double x = parseTerm();
            while (true) {
                if      (eat('+')) x += parseTerm();
                else if (eat('-')) x -= parseTerm();
                else return x;
            }
        }

        double parseTerm() {
            double x = parseFactor();
            while (true) {
                if      (eat('*')) x *= parseFactor();
                else if (eat('/')) x /= parseFactor();
                else return x;
            }
        }

        double parseFactor() {
            if (eat('+')) return parseFactor(); // unary plus
            if (eat('-')) return -parseFactor(); // unary minus

            double x;
            int startPos = this.pos;
            if (eat('(')) {
                x = parseExpression();
                eat(')');
            } else if ((ch >= '0' && ch <= '9') || ch == '.') {
                while ((ch >= '0' && ch <= '9') || ch == '.') nextChar();
                x = Double.parseDouble(expression.substring(startPos, this.pos));
            } else {
                throw new RuntimeException("Unexpected: " + (char) ch);
            }

            return x;
        }
    }.parse();
  }
}
```

Activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp">

    <EditText
        android:id="@+id/editText"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:inputType="none"
        android:textSize="24sp"
        android:gravity="end"
        android:background="#E0E0E0"
        android:padding="12dp"
        android:textColor="#000"
        android:focusable="false"
        android:cursorVisible="false"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent" />

    <GridLayout
        android:id="@+id/gridLayout"
        android:layout_width="0dp"
        android:layout_height="0dp"
        android:layout_marginTop="16dp"
        android:alignmentMode="alignMargins"
        android:columnCount="4"
        android:rowCount="5"
        app:layout_constraintTop_toBottomOf="@id/editText"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent">

        <!-- Row 1 -->
        <Button android:text="1" android:onClick="buttonClick"/>
        <Button android:text="2" android:onClick="buttonClick"/>
        <Button android:text="3" android:onClick="buttonClick"/>
        <Button android:text="+" android:onClick="buttonClick"/>
```

```
    <!-- Row 2 -->
    <Button android:text="4" android:onClick="buttonClick"/>
    <Button android:text="5" android:onClick="buttonClick"/>
    <Button android:text="6" android:onClick="buttonClick"/>
    <Button android:text="-" android:onClick="buttonClick"/>

    <!-- Row 3 -->
    <Button android:text="7" android:onClick="buttonClick"/>
    <Button android:text="8" android:onClick="buttonClick"/>
    <Button android:text="9" android:onClick="buttonClick"/>
    <Button android:text="*" android:onClick="buttonClick"/>

    <!-- Row 4 -->
    <Button android:text="." android:onClick="buttonClick"/>
    <Button android:text="0" android:onClick="buttonClick"/>
    <Button android:text="C" android:onClick="clearClick"/>
    <Button android:text="/" android:onClick="buttonClick"/>

    <!-- Row 5 -->
    <Button android:text="=" android:layout_columnSpan="4"
android:onClick="calculateClick"/>
  </GridLayout>
</androidx.constraintlayout.widget.ConstraintLayout>
```

SET B

1. **Create Android app for addition of two numbers using RelativeLayout. And perform operation.**

->

MainActivity.java

```
package com.example.relativelayoutex;


import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.view.View;
import androidx.appcompat.app.AppCompatActivity;
```

```
public class MainActivity extends AppCompatActivity {

    EditText num1, num2;
    Button btnAdd;
    TextView result;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        num1 = findViewById(R.id.num1);
        num2 = findViewById(R.id.num2);
        btnAdd = findViewById(R.id.btnAdd);
        result = findViewById(R.id.result);

        btnAdd.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String val1 = num1.getText().toString();
                String val2 = num2.getText().toString();

                if (!val1.isEmpty() && !val2.isEmpty()) {
                    int number1 = Integer.parseInt(val1);
                    int number2 = Integer.parseInt(val2);
                    int sum = number1 + number2;
                    result.setText("Result: " + sum);
                } else {
                    result.setText("Please enter both numbers");
                }
            }
        });
    }
}
```

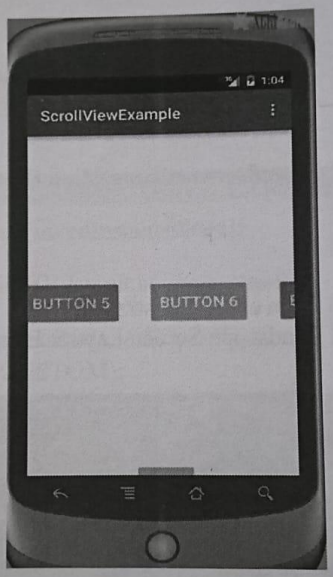
activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp">
```

```
    <EditText
        android:id="@+id/num1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter first number"
        android:inputType="number" />

    <EditText
        android:id="@+id/num2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter second number"
        android:layout_below="@id/num1"
        android:layout_marginTop="16dp"
        android:inputType="number" />

    <Button
        android:id="@+id/btnAdd"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Add"
        android:layout_below="@id/num2"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="20dp"/>

    <TextView
        android:id="@+id/result"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Result: "
        android:textSize="18sp"
        android:layout_below="@id/btnAdd"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="24dp"/>
</RelativeLayout>
```

---

2. Using ScrollView create application to scroll the buttons in horizontal direction as shown below.



->

MainActivity.java

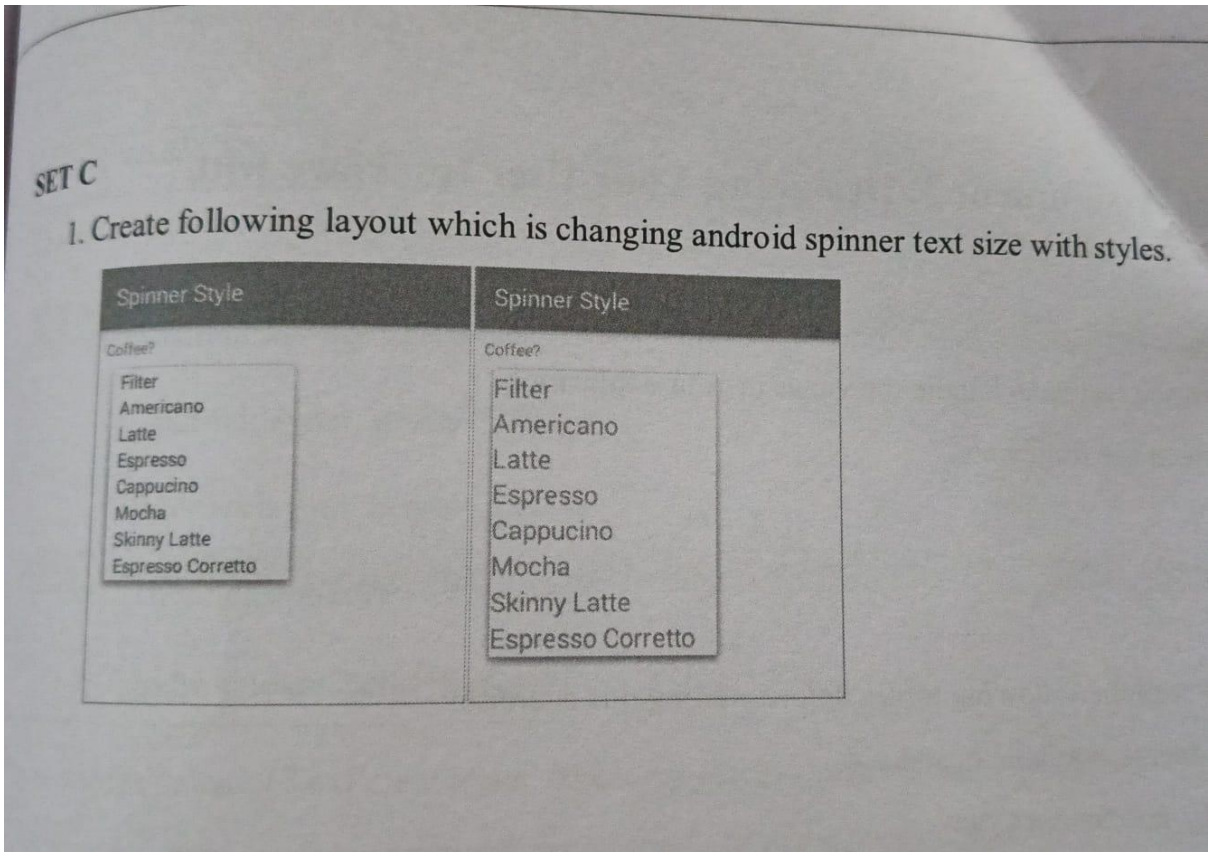
```
package com.example.horizontalscrollviewex;

import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -
> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
```

```
        return insets;
    });
  }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent">

  <HorizontalScrollView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_centerInParent="true">

    <LinearLayout
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:orientation="horizontal"
      android:padding="16dp">

      <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Button 1" />

      <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Button 2"
        android:layout_marginStart="8dp" />

      <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Button 3"
        android:layout_marginStart="8dp" />

      <Button
        android:layout_width="wrap_content"
```

```
            android:layout_height="wrap_content"
            android:text="Button 4"
            android:layout_marginStart="8dp" />

        <Button
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Button 5"
            android:layout_marginStart="8dp" />

        <Button
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Button 6"
            android:layout_marginStart="8dp" />
    </LinearLayout>
  </HorizontalScrollView>
</RelativeLayout>
```



->

MainActivity.java

```
package com.example.spinnerex;



import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.Spinner;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

   String[] coffeeOptions = {
         "Filter", "Americano", "Latte", "Espresso",
         "Cappucino", "Mocha", "Skinny Latte", "Espresso Corretto"
   };

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);

      Spinner coffeeSpinner = findViewById(R.id.coffeeSpinner);

      ArrayAdapter<String> adapter = new ArrayAdapter<>(
            this,
            R.layout.spinner_item, // custom layout
            coffeeOptions
      );

      adapter.setDropDownViewResource(R.layout.spinner_item);
      coffeeSpinner.setAdapter(adapter);
   }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
```

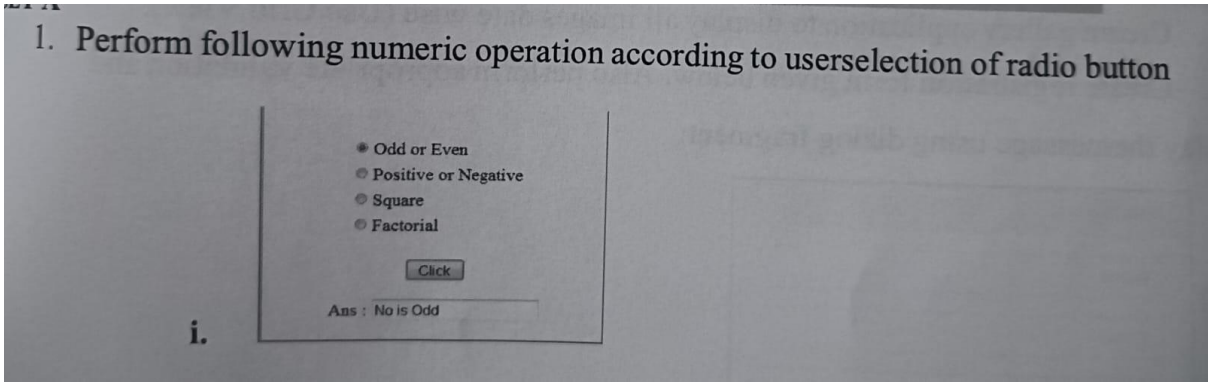
```
    android:layout_height="match_parent"
    android:padding="16dp">

    <TextView
        android:id="@+id/spinnerLabel"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Coffee?"
        android:textSize="18sp"
        android:textStyle="bold" />

    <Spinner
        android:id="@+id/coffeeSpinner"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_below="@id/spinnerLabel"
        android:layout_marginTop="8dp"
        android:spinnerMode="dropdown"/>
</RelativeLayout>
```

Spinner_item.xml

```
<?xml version="1.0" encoding="utf-8"?>
<TextView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:textSize="20sp"
    android:padding="8dp"
    android:textColor="#000000" />
```

ASSIGNMENT NO.3]

SET-A]



->

MainActivity.java

```
package com.example.simpleandroidapp;


import android.os.Bundle;
import android.view.View;
import android.widget.*;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText inputNumber;
    RadioGroup radioGroup;
    RadioButton rbOddEven, rbPosNeg, rbSquare, rbFactorial;
    Button btnCheck;
    TextView txtResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        inputNumber = findViewById(R.id.inputNumber);
        radioGroup = findViewById(R.id.radioGroup);
        rbOddEven = findViewById(R.id.rbOddEven);
        rbPosNeg = findViewById(R.id.rbPosNeg);
        rbSquare = findViewById(R.id.rbSquare);
        rbFactorial = findViewById(R.id.rbFactorial);
        btnCheck = findViewById(R.id.btnCheck);
        txtResult = findViewById(R.id.txtResult);
```

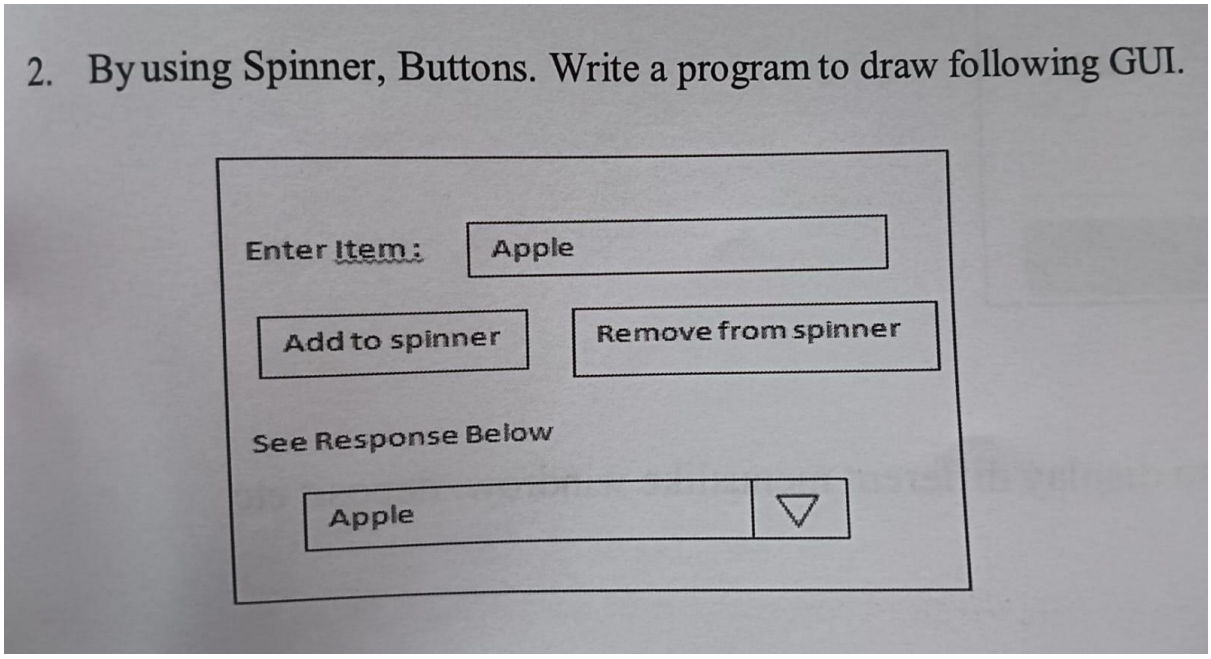
```
        btnCheck.setOnClickListener(v -> {
            String inputStr = inputNumber.getText().toString();
            if (inputStr.isEmpty()) {
                txtResult.setText("Please enter a number.");
                return;
            }

            int num = Integer.parseInt(inputStr);
            String result = "";

            if (rbOddEven.isChecked()) {
                result = (num % 2 == 0) ? "No is Even" : "No is Odd";
            } else if (rbPosNeg.isChecked()) {
                result = (num >= 0) ? "No is Positive" : "No is Negative";
            } else if (rbSquare.isChecked()) {
                result = "Square is " + (num * num);
            } else if (rbFactorial.isChecked()) {
                int fact = 1;
                for (int i = 1; i <= num; i++) {
                    fact *= i;
                }
                result = "Factorial is " + fact;
            } else {
                result = "Please select an operation.";
            }

            txtResult.setText("Ans: " + result);
        });
    }
}
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <EditText
        android:id="@+id/inputNumber"
        android:layout_width="match_parent"
```

```
    android:layout_height="wrap_content"
    android:hint="Enter a number"
    android:inputType="number" />

<RadioGroup
    android:id="@+id/radioGroup"
    android:layout_width="match_parent"
    android:layout_height="wrap_content">

    <RadioButton
        android:id="@+id/rbOddEven"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Odd or Even" />

    <RadioButton
        android:id="@+id/rbPosNeg"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Positive or Negative" />

    <RadioButton
        android:id="@+id/rbSquare"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Square" />

    <RadioButton
        android:id="@+id/rbFactorial"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Factorial" />
</RadioGroup>

<Button
    android:id="@+id/btnCheck"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Click" />

<TextView
    android:id="@+id/txtResult"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
```

```
        android:paddingTop="20dp"
        android:text="Ans: "
        android:textSize="16sp" />
    </LinearLayout>
```



->**MainActivity.java**

```
package com.example.spinnerexample;

import android.os.Bundle;
import android.view.View;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;
import java.util.ArrayList;

public class MainActivity extends AppCompatActivity {

    EditText editTextItem;
    Button btnAdd, btnRemove;
    Spinner spinner;
    ArrayAdapter<String> adapter;
    ArrayList<String> itemsList;

    @Override
```

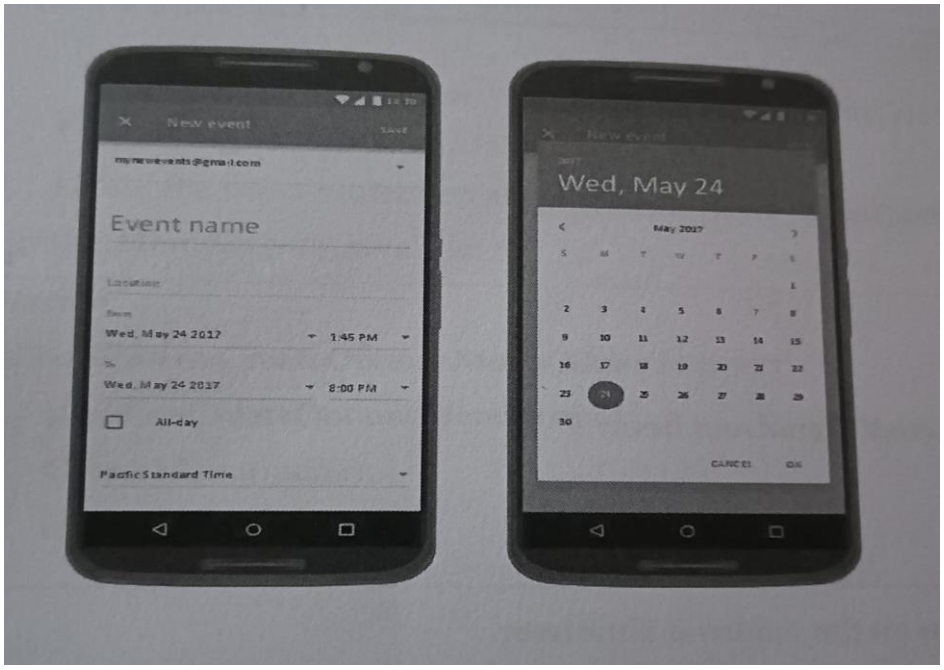
```
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        editTextItem = findViewById(R.id.editTextItem);
        btnAdd = findViewById(R.id.btnAdd);
        btnRemove = findViewById(R.id.btnRemove);
        spinner = findViewById(R.id.spinner);

        itemsList = new ArrayList<>();
        itemsList.add("Apple");

        adapter = new ArrayAdapter<>(this, android.R.layout.simple_spinner_dropdown_item,
itemsList);
        spinner.setAdapter(adapter);

        btnAdd.setOnClickListener(view -> {
            String newItem = editTextItem.getText().toString().trim();
            if (!newItem.isEmpty() && !itemsList.contains(newItem)) {
                itemsList.add(newItem);
                adapter.notifyDataSetChanged();
                editTextItem.setText("");
            }
        });

        btnRemove.setOnClickListener(view -> {
            String itemToRemove = editTextItem.getText().toString().trim();
            if (itemsList.contains(itemToRemove)) {
                itemsList.remove(itemToRemove);
                adapter.notifyDataSetChanged();
                editTextItem.setText("");
            }
        });
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="20dp">
```

```
    <TextView
        android:text="Enter Item:"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content" />

    <EditText
        android:id="@+id/editTextItem"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter item here" />

    <LinearLayout
        android:orientation="horizontal"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="10dp">

        <Button
            android:id="@+id/btnAdd"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Add to spinner" />

        <Button
            android:id="@+id/btnRemove"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Remove from spinner"
            android:layout_marginStart="10dp"/>
    </LinearLayout>

    <TextView
        android:text="See Response Below"
        android:layout_marginTop="15dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content" />

    <Spinner
        android:id="@+id/spinner"
        android:layout_width="match_parent"
        android:layout_height="wrap_content" />
</LinearLayout>
```

3.Create application to demonstrate date and time picker.



->

MainActivity.java

```
package com.example.simpleandroidapplication;

import android.app.DatePickerDialog;
import android.app.TimePickerDialog;
import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

import com.example.simpleandroidapplication.R;

import java.util.Calendar;

public class MainActivity extends AppCompatActivity {

    Button btnStartDate, btnStartTime, btnEndDate, btnEndTime;
    CheckBox checkBoxAllDay;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnStartDate = findViewById(R.id.btnStartDate);
```

```
        btnStartTime = findViewById(R.id.btnStartTime);
        btnEndDate = findViewById(R.id.btnEndDate);
        btnEndTime = findViewById(R.id.btnEndTime);
        checkBoxAllDay = findViewById(R.id.checkBoxAllDay);

        btnStartDate.setOnClickListener(v -> showDatePicker(btnStartDate));
        btnEndDate.setOnClickListener(v -> showDatePicker(btnEndDate));
        btnStartTime.setOnClickListener(v -> showTimePicker(btnStartTime));
        btnEndTime.setOnClickListener(v -> showTimePicker(btnEndTime));

        checkBoxAllDay.setOnCheckedChangeListener((buttonView, isChecked) -> {
            btnStartTime.setEnabled(!isChecked);
            btnEndTime.setEnabled(!isChecked);
        });
    }

    private void showDatePicker(Button button) {
        final Calendar c = Calendar.getInstance();
        int year = c.get(Calendar.YEAR);
        int month = c.get(Calendar.MONTH);
        int day = c.get(Calendar.DAY_OF_MONTH);

        DatePickerDialog datePickerDialog = new DatePickerDialog(this,
                (view, selectedYear, selectedMonth, selectedDay) -> {
                    String date = selectedDay + "/" + (selectedMonth + 1) + "/" + selectedYear;
                    button.setText(date);
                }, year, month, day);
        datePickerDialog.show();
    }

    private void showTimePicker(Button button) {
        final Calendar c = Calendar.getInstance();
        int hour = c.get(Calendar.HOUR_OF_DAY);
        int minute = c.get(Calendar.MINUTE);

        TimePickerDialog timePickerDialog = new TimePickerDialog(this,
                (view, selectedHour, selectedMinute) -> {
                    String time = String.format("%02d:%02d", selectedHour, selectedMinute);
                    button.setText(time);
                }, hour, minute, false);
        timePickerDialog.show();
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <EditText
        android:id="@+id/editTextEvent"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Event name" />

    <EditText
        android:id="@+id/editTextLocation"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Location" />

    <TextView
        android:text="From"
        android:layout_marginTop="10dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content" />

    <LinearLayout
        android:orientation="horizontal"
        android:layout_width="match_parent"
        android:layout_height="wrap_content">

        <Button
            android:id="@+id/btnStartDate"
            android:layout_width="0dp"
            android:layout_weight="1"
            android:text="Select Date"
            android:layout_height="wrap_content" />

        <Button
            android:id="@+id/btnStartTime"
            android:layout_width="0dp"
            android:layout_weight="1"
```

```
        android:text="Select Time"
        android:layout_height="wrap_content" />
    </LinearLayout>

    <TextView
        android:text="To"
        android:layout_marginTop="10dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content" />

    <LinearLayout
        android:orientation="horizontal"
        android:layout_width="match_parent"
        android:layout_height="wrap_content">

        <Button
            android:id="@+id/btnEndDate"
            android:layout_width="0dp"
            android:layout_weight="1"
            android:text="Select Date"
            android:layout_height="wrap_content" />

        <Button
            android:id="@+id/btnEndTime"
            android:layout_width="0dp"
            android:layout_weight="1"
            android:text="Select Time"
            android:layout_height="wrap_content" />
    </LinearLayout>

    <CheckBox
        android:id="@+id/checkBoxAllDay"
        android:layout_marginTop="10dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="All-day" />
</LinearLayout>
```

SET B

1. **Construct an app that toggles a light bulb on and off when the user clicks on togglebutton.**

->

MainActivity.java

```
package com.example.lightbulbimg;

import android.os.Bundle;
import android.widget.ImageView;
import android.widget.ToggleButton;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private ToggleButton toggleButton;
    private ImageView bulbImage;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        toggleButton = findViewById(R.id.toggleButton);
        bulbImage = findViewById(R.id.imageViewBulb);

        toggleButton.setOnCheckedChangeListener((buttonView, isChecked) -> {
            if (isChecked) {
                bulbImage.setImageResource(R.drawable.bulb_on);
            } else {
                bulbImage.setImageResource(R.drawable.bulb_off);
            }
        });
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
```

```
    android:padding="24dp">

    <ImageView
        android:id="@+id/imageViewBulb"
        android:layout_width="200dp"
        android:layout_height="200dp"
        android:src="@drawable/bulb_off"
        android:contentDescription="Light bulb" />

    <ToggleButton
        android:id="@+id/toggleButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textOn="Turn Off"
        android:textOff="Turn On"
        android:layout_marginTop="24dp"/>
</LinearLayout>
```

---

2. Create gallery application to display all images date wise (Use Grid View).

->

<u>AndroidManifest.xml</u>

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.example.gallaryapp">

    <application
        android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.AppCompat.Light.DarkActionBar">

        <!-- Main Activity with explicit android:exported -->
        <activity android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
```

```
            <category android:name="android.intent.category.LAUNCHER" />
        </intent-filter>
    </activity>
  </application>

  <!-- Permission to read external storage -->
  <uses-permission android:name="android.permission.READ_EXTERNAL_STORAGE" />
</manifest>
```

ImageAdapter.java

```
package com.example.gallaryapp;

import android.content.Context;
import android.graphics.Bitmap;
import android.graphics.BitmapFactory;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.BaseAdapter;
import android.widget.GridView;
import android.widget.ImageView;

import java.util.ArrayList;

public class ImageAdapter extends BaseAdapter {

  private Context context;
  private ArrayList<ImageItem> imageList;

  public ImageAdapter(Context context, ArrayList<ImageItem> imageList) {
    this.context = context;
    this.imageList = imageList;
  }

  @Override
  public int getCount() {
    return imageList.size();
  }

  @Override
  public Object getItem(int position) {
    return imageList.get(position);
```

```
    }

    @Override
    public long getItemId(int position) {
        return position;
    }

    @Override
    public View getView(int position, View convertView, ViewGroup parent) {
        ImageView imageView;

        if (convertView == null) {
            imageView = new ImageView(context);
            imageView.setLayoutParams(new GridView.LayoutParams(300, 300));
            imageView.setScaleType(ImageView.ScaleType.CENTER_CROP);
        } else {
            imageView = (ImageView) convertView;
        }

        // Get the image path and decode it into a bitmap
        String imagePath = imageList.get(position).getImagePath();
        Bitmap bitmap = BitmapFactory.decodeFile(imagePath);
        imageView.setImageBitmap(bitmap);

        return imageView;
    }
}
```

MainActivity.java

```
package com.example.gallaryapp;


import android.Manifest;
import android.content.ContentResolver;
import android.content.pm.PackageManager;
import android.database.Cursor;
import android.net.Uri;
import android.os.Bundle;
import android.provider.MediaStore;
import android.widget.GridView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
```

```
import androidx.core.content.ContextCompat;

import java.util.ArrayList;

public class MainActivity extends AppCompatActivity {

    private static final int REQUEST_PERMISSION = 1;
    private GridView gridView;
    private ArrayList<ImageItem> imageList;
    private ImageAdapter imageAdapter;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        gridView = findViewById(R.id.gridView);
        imageList = new ArrayList<>();

        // Check if permission is granted
        if (ContextCompat.checkSelfPermission(this,
Manifest.permission.READ_EXTERNAL_STORAGE) !=
PackageManager.PERMISSION_GRANTED) {
            ActivityCompat.requestPermissions(this, new
String[]{Manifest.permission.READ_EXTERNAL_STORAGE}, REQUEST_PERMISSION);
        } else {
            loadImages();
        }
    }

    // Handle permission result
    @Override
    public void onRequestPermissionsResult(int requestCode, String[] permissions, int[]
grantResults) {
        if (requestCode == REQUEST_PERMISSION) {
            if (grantResults.length > 0 && grantResults[0] ==
PackageManager.PERMISSION_GRANTED) {
                loadImages();
            } else {
                Toast.makeText(this, "Permission denied", Toast.LENGTH_SHORT).show();
            }
        }
    }
```

```
    // Load images from external storage
    private void loadImages() {
        ContentResolver contentResolver = getContentResolver();
        Uri imageUri = MediaStore.Images.Media.EXTERNAL_CONTENT_URI;

        // Query to get all image data (ID, path, date taken)
        Cursor cursor = contentResolver.query(imageUri, null, null, null,
MediaStore.Images.Media.DATE_TAKEN + " DESC");

        if (cursor != null && cursor.moveToFirst()) {
            int idColumn = cursor.getColumnIndex(MediaStore.Images.Media._ID);
            int pathColumn = cursor.getColumnIndex(MediaStore.Images.Media.DATA);
            int dateColumn =
cursor.getColumnIndex(MediaStore.Images.Media.DATE_TAKEN);

            do {
                String imagePath = cursor.getString(pathColumn);
                long dateTaken = cursor.getLong(dateColumn);

                // Add images to the list
                imageList.add(new ImageItem(imagePath, dateTaken));

            } while (cursor.moveToNext());
            cursor.close();

            // Set the adapter for GridView
            imageAdapter = new ImageAdapter(this, imageList);
            gridView.setAdapter(imageAdapter);
        }
    }
}
```

ImageItem

```
package com.example.gallaryapp;


public class ImageItem {
    private String imagePath;
    private long dateTaken;

    public ImageItem(String imagePath, long dateTaken) {
        this.imagePath = imagePath;
        this.dateTaken = dateTaken;
    }
```

```
    public String getImagePath() {
        return imagePath;
    }

    public long getDateTaken() {
        return dateTaken;
    }
}
```

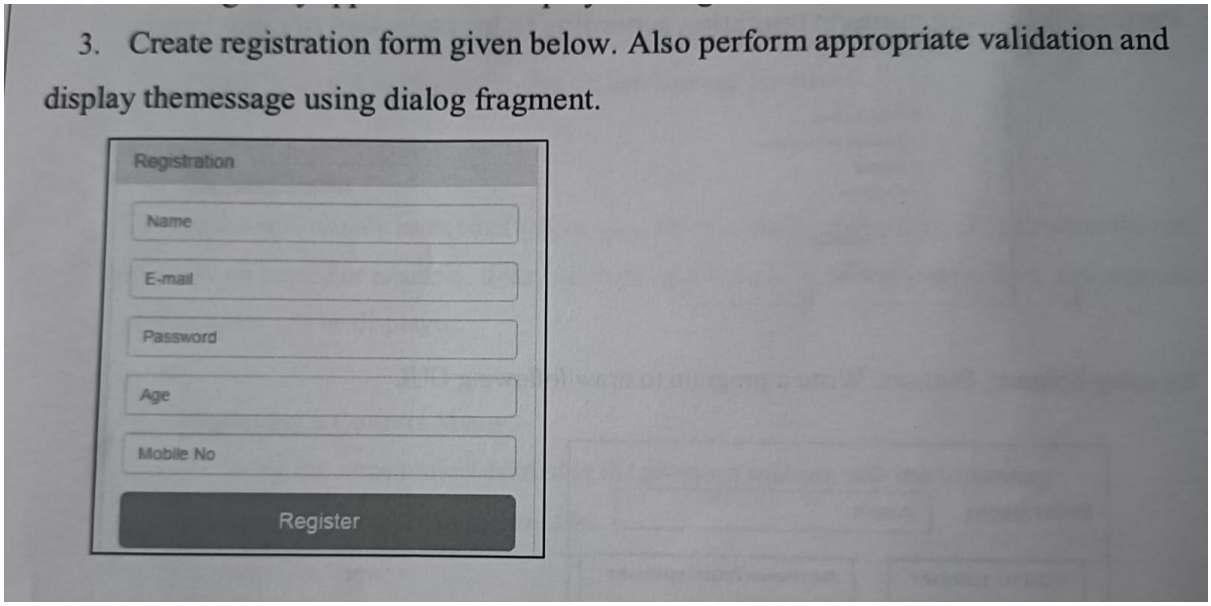
activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <!-- GridView to display images -->
    <GridView
        android:id="@+id/gridView"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:numColumns="3"
        android:columnWidth="100dp"
        android:horizontalSpacing="10dp"
        android:verticalSpacing="10dp"
        android:stretchMode="columnWidth" />
</LinearLayout>
```

item_image.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.cardview.widget.CardView
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_margin="8dp"
    android:radius="8dp">

    <ImageView
        android:id="@+id/imageView"
        android:layout_width="100dp"
```

```
        android:layout_height="100dp"
        android:layout_gravity="center"
        android:scaleType="centerCrop"
        android:layout_margin="4dp"
        android:contentDescription="Gallery Image" />
</androidx.cardview.widget.CardView>
```



->

MainActivity.java

```
package com.example.validationform;


import android.os.Bundle;
import android.text.TextUtils;
import android.util.Patterns;
import android.view.View;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText etName, etEmail, etPassword, etAge, etMobile;
    Button btnRegister;

    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etName = findViewById(R.id.etName);
        etEmail = findViewById(R.id.etEmail);
        etPassword = findViewById(R.id.etPassword);
        etAge = findViewById(R.id.etAge);
        etMobile = findViewById(R.id.etMobile);
        btnRegister = findViewById(R.id.btnRegister);

        btnRegister.setOnClickListener(v -> {
            if (validate()) {
                String info = "Name: " + etName.getText().toString() + "\n"
                        + "Email: " + etEmail.getText().toString() + "\n"
                        + "Age: " + etAge.getText().toString() + "\n"
                        + "Mobile: " + etMobile.getText().toString();

                SuccessDialogFragment dialog =
SuccessDialogFragment.newInstance("Registration Successful!", info);
                dialog.show(getSupportFragmentManager(), "successDialog");
            }
        });
    }

    private boolean validate() {
        if (TextUtils.isEmpty(etName.getText())) {
            etName.setError("Name required");
            return false;
        }

        if (TextUtils.isEmpty(etEmail.getText()) ||
!Patterns.EMAIL_ADDRESS.matcher(etEmail.getText()).matches()) {
            etEmail.setError("Valid email required");
            return false;
        }

        if (TextUtils.isEmpty(etPassword.getText()) || etPassword.length() < 6) {
            etPassword.setError("Password must be 6+ chars");
            return false;
        }

        if (TextUtils.isEmpty(etAge.getText())) {
            etAge.setError("Age required");
```

```
            return false;
        }

        if (TextUtils.isEmpty(etMobile.getText()) || etMobile.length() != 10) {
            etMobile.setError("Valid 10-digit mobile no required");
            return false;
        }

        return true;
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical" android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp">

    <EditText
        android:id="@+id/etName"
        android:hint="Name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/etEmail"
        android:hint="E-mail"
        android:inputType="textEmailAddress"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/etPassword"
        android:hint="Password"
        android:inputType="textPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/etAge"
        android:hint="Age"
```

```
        android:inputType="number"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/etMobile"
        android:hint="Mobile No"
        android:inputType="phone"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/btnRegister"
        android:text="Register"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
</LinearLayout>
```

SucessDialogFragment.java

```
package com.example.validationform;

import android.app.Dialog;
import android.os.Bundle;
import androidx.annotation.NonNull;
import androidx.fragment.app.DialogFragment;
import androidx.appcompat.app.AlertDialog;

public class SuccessDialogFragment extends DialogFragment {

    public static SuccessDialogFragment newInstance(String title, String message) {
        SuccessDialogFragment frag = new SuccessDialogFragment();
        Bundle args = new Bundle();
        args.putString("title", title);
        args.putString("message", message);
        frag.setArguments(args);
        return frag;
    }

    @NonNull
    @Override
    public Dialog onCreateDialog(Bundle savedInstanceState) {
        String title = getArguments().getString("title", "Success");
```

```
        String message = getArguments().getString("message", "");

        return new AlertDialog.Builder(requireActivity())
            .setTitle(title)
            .setMessage(message)
            .setPositiveButton("OK", (dialog, which) -> dismiss())
            .create();
    }
}
```

SET C

1. Construct a bank app to display different menuslike windrow, deposit etc.

->

MainActivity.java

```
package com.example.menuslike;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private double balance = 0.0;
    private TextView balanceText;
    private EditText amountInput;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        balanceText = findViewById(R.id.balanceText);
        amountInput = findViewById(R.id.amountInput);
```

```
    Button depositButton = findViewById(R.id.depositButton);
    Button withdrawButton = findViewById(R.id.withdrawButton);
    Button checkBalanceButton = findViewById(R.id.checkBalanceButton);

    // Deposit Button
    depositButton.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            String inputAmount = amountInput.getText().toString();
            if (!inputAmount.isEmpty()) {
                double depositAmount = Double.parseDouble(inputAmount);
                if (depositAmount > 0) {
                    balance += depositAmount;
                    updateBalance();
                }
            }
        }
    });

    // Withdraw Button
    withdrawButton.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            String inputAmount = amountInput.getText().toString();
            if (!inputAmount.isEmpty()) {
                double withdrawAmount = Double.parseDouble(inputAmount);
                if (withdrawAmount > 0 && withdrawAmount <= balance) {
                    balance -= withdrawAmount;
                    updateBalance();
                } else {
                    // Handle insufficient balance
                }
            }
        }
    });

    // Check Balance Button
    checkBalanceButton.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            updateBalance();
        }
    });
}
```

```
    // Update the displayed balance
    private void updateBalance() {
        balanceText.setText("Balance: $" + balance);
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <!-- Balance Display -->
    <TextView
        android:id="@+id/balanceText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Balance: $0.0"
        android:textSize="24sp"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="50dp" />

    <!-- Deposit Button -->
    <Button
        android:id="@+id/depositButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Deposit"
        android:layout_below="@id/balanceText"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="30dp" />

    <!-- Withdraw Button -->
    <Button
        android:id="@+id/withdrawButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Withdraw"
        android:layout_below="@id/depositButton"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="20dp" />
```

```
    <!-- Balance Button -->
    <Button
        android:id="@+id/checkBalanceButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Check Balance"
        android:layout_below="@id/withdrawButton"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="20dp" />

    <!-- Amount Input -->
    <EditText
        android:id="@+id/amountInput"
        android:layout_width="200dp"
        android:layout_height="wrap_content"
        android:hint="Enter amount"
        android:inputType="numberDecimal"
        android:layout_below="@id/checkBalanceButton"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="30dp" />
</RelativeLayout>
```

ASSIGNMENT No.4]

SET-A]

1. Create table Company (id, name, address, phno). Create Application for Performing the following operation on the table.

i) Insert New Company Details.

ii) Show All the Company Details.

->

MainActivity.java

```
package com.example.companyapplication;

import android.database.Cursor;
import android.os.Bundle;
```

```
import android.view.View;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    CompanyDatabaseHelper dbHelper;
    EditText idInput, nameInput, addressInput, phnoInput;
    TextView resultView;
    Button insertBtn, showBtn;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        dbHelper = new CompanyDatabaseHelper(this);

        idInput = findViewById(R.id.idInput);
        nameInput = findViewById(R.id.nameInput);
        addressInput = findViewById(R.id.addressInput);
        phnoInput = findViewById(R.id.phnoInput);
        resultView = findViewById(R.id.resultView);

        insertBtn = findViewById(R.id.insertBtn);
        showBtn = findViewById(R.id.showBtn);

        insertBtn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                int id = Integer.parseInt(idInput.getText().toString());
                String name = nameInput.getText().toString();
                String address = addressInput.getText().toString();
                String phno = phnoInput.getText().toString();

                boolean inserted = dbHelper.insertCompany(id, name, address, phno);
                Toast.makeText(MainActivity.this,
                        inserted ? "Inserted Successfully" : "Insert Failed",
                        Toast.LENGTH_SHORT).show();
            }
        });

        showBtn.setOnClickListener(new View.OnClickListener() {
            @Override
```

```
        public void onClick(View v) {
            Cursor cursor = dbHelper.getAllCompanies();
            if (cursor.getCount() == 0) {
                resultView.setText("No records found.");
                return;
            }

            StringBuilder sb = new StringBuilder();
            while (cursor.moveToNext()) {
                sb.append("ID: ").append(cursor.getInt(0)).append("\n")
                        .append("Name: ").append(cursor.getString(1)).append("\n")
                        .append("Address: ").append(cursor.getString(2)).append("\n")
                        .append("Phone: ").append(cursor.getString(3)).append("\n\n");
            }

            resultView.setText(sb.toString());
        }
    });
  }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:orientation="vertical"
        android:padding="20dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content">

        <EditText android:id="@+id/idInput"
            android:hint="ID"
            android:inputType="number"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>

        <EditText android:id="@+id/nameInput"
            android:hint="Company Name"
            android:layout_width="match_parent"
```

```
        android:layout_height="wrap_content"/>

    <EditText android:id="@+id/addressInput"
        android:hint="Address"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText android:id="@+id/phnoInput"
        android:hint="Phone Number"
        android:inputType="phone"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button android:id="@+id/insertBtn"
        android:text="Insert Company"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="10dp"/>

    <Button android:id="@+id/showBtn"
        android:text="Show All Companies"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="10dp"/>

    <TextView android:id="@+id/resultView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="16sp"
        android:paddingTop="20dp"/>
  </LinearLayout>
</ScrollView>
```

CompanyDatabaseHelper

```
package com.example.companyapplication;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.*;

public class CompanyDatabaseHelper extends SQLiteOpenHelper {
```

```
private static final String DB_NAME = "CompanyDB";
private static final int DB_VERSION = 1;

public static final String TABLE_NAME = "Company";
public static final String COL_ID = "id";
public static final String COL_NAME = "name";
public static final String COL_ADDRESS = "address";
public static final String COL_PHNO = "phno";

public CompanyDatabaseHelper(Context context) {
   super(context, DB_NAME, null, DB_VERSION);
}

@Override
public void onCreate(SQLiteDatabase db) {
   String createTable = "CREATE TABLE " + TABLE_NAME + " (" +
         COL_ID + " INTEGER PRIMARY KEY, " +
         COL_NAME + " TEXT, " +
         COL_ADDRESS + " TEXT, " +
         COL_PHNO + " TEXT)";
   db.execSQL(createTable);
}

@Override
public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
   db.execSQL("DROP TABLE IF EXISTS " + TABLE_NAME);
   onCreate(db);
}

public boolean insertCompany(int id, String name, String address, String phno) {
   SQLiteDatabase db = this.getWritableDatabase();
   ContentValues cv = new ContentValues();
   cv.put(COL_ID, id);
   cv.put(COL_NAME, name);
   cv.put(COL_ADDRESS, address);
   cv.put(COL_PHNO, phno);

   long result = db.insert(TABLE_NAME, null, cv);
   return result != -1;
}

public Cursor getAllCompanies() {
   SQLiteDatabase db = this.getReadableDatabase();
```

```
        return db.rawQuery("SELECT * FROM " + TABLE_NAME, null);
    }
}
```

---

2. Create table Student (sno, s_name,s_class,s_addr)

Teacher (tno, t_name, qualification, experience) Student-Teacher has many to many relationship.

Using above database Write Application to accept a teacher name from user and display the names of students along with subjects to whom teacher is teaching.

->

MainActivity.java

```
package com.example.teacherstudapp;


import android.database.Cursor;
import android.os.Bundle;
import android.view.View;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText teacherNameInput;
    Button searchBtn;
    TextView resultView;
    DatabaseHelper db;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        teacherNameInput = findViewById(R.id.teacherNameInput);
        searchBtn = findViewById(R.id.searchBtn);
        resultView = findViewById(R.id.resultView);
        db = new DatabaseHelper(this);
```

```
        searchBtn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String teacherName = teacherNameInput.getText().toString().trim();
                if (teacherName.isEmpty()) {
                    resultView.setText("Please enter a teacher name.");
                    return;
                }

                Cursor cursor = db.getStudentsByTeacherName(teacherName);
                if (cursor.getCount() == 0) {
                    resultView.setText("No students found for this teacher.");
                    return;
                }

                StringBuilder sb = new StringBuilder();
                while (cursor.moveToNext()) {
                    sb.append("Student: ").append(cursor.getString(0))
                            .append("\nSubject: ").append(cursor.getString(1)).append("\n\n");
                }

                resultView.setText(sb.toString());
            }
        });
    }
}
```

activity_main.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:padding="16dp"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <EditText
        android:id="@+id/teacherNameInput"
        android:hint="Enter Teacher Name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content" />

    <Button
```

```
        android:id="@+id/searchBtn"
        android:text="Search Students"
        android:layout_width="match_parent"
        android:layout_height="wrap_content" />

    <TextView
        android:id="@+id/resultView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="16sp"
        android:layout_marginTop="16dp" />
</LinearLayout>
```

DatabaseHelper.java

```
package com.example.teacherstudapp;


import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.*;

public class DatabaseHelper extends SQLiteOpenHelper {

    public DatabaseHelper(Context context) {
        super(context, "TeachStudentDB", null, 1);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        db.execSQL("CREATE TABLE Student(sno INTEGER PRIMARY KEY, s_name TEXT,
s_class TEXT, s_addr TEXT)");
        db.execSQL("CREATE TABLE Teacher(tno INTEGER PRIMARY KEY, t_name TEXT,
qualification TEXT, experience INTEGER)");
        db.execSQL("CREATE TABLE StudentTeacher(id INTEGER PRIMARY KEY
AUTOINCREMENT, sno INTEGER, tno INTEGER, subject TEXT, " +
                "FOREIGN KEY(sno) REFERENCES Student(sno), FOREIGN KEY(tno) REFERENCES
Teacher(tno))");

        // Sample data for demo
        db.execSQL("INSERT INTO Student VALUES (1, 'Manisha', 'BCSA', 'NY'), (2, 'Tanu',
'BCSB', 'LA')");
        db.execSQL("INSERT INTO Teacher VALUES (101, 'Mr. Gobare sir', 'MSc', 5), (102,
```

```
'Ms. Danave Mam', 'MA', 4)");
        db.execSQL("INSERT INTO StudentTeacher (sno, tno, subject) VALUES (1, 101, 'SQL'),
(2, 101, 'Software Testing'), (1, 102, 'MongoDB')");
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
        db.execSQL("DROP TABLE IF EXISTS StudentTeacher");
        db.execSQL("DROP TABLE IF EXISTS Student");
        db.execSQL("DROP TABLE IF EXISTS Teacher");
        onCreate(db);
    }

    public Cursor getStudentsByTeacherName(String t_name) {
        SQLiteDatabase db = this.getReadableDatabase();
        return db.rawQuery(
                "SELECT s.s_name, st.subject " +
                        "FROM Student s " +
                        "JOIN StudentTeacher st ON s.sno = st.sno " +
                        "JOIN Teacher t ON t.tno = st.tno " +
                        "WHERE t.t_name = ?",
                new String[]{t_name});
    }
}
```

---

3. Create Following Table:

Emp (emp_no,emp_name, address, phone, salary)

Dept (dept_no,dept_name, location)

Emp-Dept is related with one-many relationship.

Create application for performing the following Operation on the table

1) Add Records into Emp and Dept table.

2) Accept Department name from User and delete employee information which belongs to that department.

->

MainActivity.java

```
package com.example.empdeptapp;

import android.os.Bundle;
import android.view.View;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    DatabaseHelper db;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        db = new DatabaseHelper(this);

        EditText deptNo = findViewById(R.id.deptNo);
        EditText deptName = findViewById(R.id.deptName);
        EditText location = findViewById(R.id.location);

        EditText empNo = findViewById(R.id.empNo);
        EditText empName = findViewById(R.id.empName);
        EditText address = findViewById(R.id.address);
        EditText phone = findViewById(R.id.phone);
        EditText salary = findViewById(R.id.salary);
        EditText empDeptNo = findViewById(R.id.empDeptNo);

        EditText deleteDeptName = findViewById(R.id.deleteDeptName);
        TextView resultText = findViewById(R.id.resultText);

        Button addDeptBtn = findViewById(R.id.addDeptBtn);
        Button addEmpBtn = findViewById(R.id.addEmpBtn);
        Button deleteEmpBtn = findViewById(R.id.deleteEmpBtn);

        addDeptBtn.setOnClickListener(v -> {
            try {
                int no = Integer.parseInt(deptNo.getText().toString());
                String name = deptName.getText().toString();
                String loc = location.getText().toString();
```

```
            db.insertDept(no, name, loc);
            resultText.setText("✅ Department added.");
        } catch (Exception e) {
            resultText.setText("❌ Error adding department.");
        }
    });

    addEmpBtn.setOnClickListener(v -> {
        try {
            int no = Integer.parseInt(empNo.getText().toString());
            String name = empName.getText().toString();
            String addr = address.getText().toString();
            String ph = phone.getText().toString();
            double sal = Double.parseDouble(salary.getText().toString());
            int dept = Integer.parseInt(empDeptNo.getText().toString());
            db.insertEmp(no, name, addr, ph, sal, dept);
            resultText.setText("✅ Employee added.");
        } catch (Exception e) {
            resultText.setText("❌ Error adding employee.");
        }
    });

    deleteEmpBtn.setOnClickListener(v -> {
        String dept = deleteDeptName.getText().toString();
        int deleted = db.deleteEmployeesByDeptName(dept);
        resultText.setText("🗑️ Deleted " + deleted + " employees from department '" +
dept + "'");
    });
  }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:padding="16dp">

  <LinearLayout android:orientation="vertical" android:layout_width="match_parent"
android:layout_height="wrap_content">

    <TextView android:text="Add Department" android:textStyle="bold"
```

```
android:textSize="18sp" />
        <EditText android:id="@+id/deptNo" android:hint="Department No"
android:inputType="number" />
        <EditText android:id="@+id/deptName" android:hint="Department Name" />
        <EditText android:id="@+id/location" android:hint="Location" />
        <Button android:id="@+id/addDeptBtn" android:text="Add Department" />

        <View android:layout_height="20dp" />

        <TextView android:text="Add Employee" android:textStyle="bold"
android:textSize="18sp" />
        <EditText android:id="@+id/empNo" android:hint="Employee No"
android:inputType="number" />
        <EditText android:id="@+id/empName" android:hint="Employee Name" />
        <EditText android:id="@+id/address" android:hint="Address" />
        <EditText android:id="@+id/phone" android:hint="Phone" />
        <EditText android:id="@+id/salary" android:hint="Salary"
android:inputType="numberDecimal" />
        <EditText android:id="@+id/empDeptNo" android:hint="Department No"
android:inputType="number" />
        <Button android:id="@+id/addEmpBtn" android:text="Add Employee" />

        <View android:layout_height="20dp" />

        <TextView android:text="Delete Employees by Dept Name" android:textStyle="bold"
android:textSize="18sp" />
        <EditText android:id="@+id/deleteDeptName" android:hint="Department Name" />
        <Button android:id="@+id/deleteEmpBtn" android:text="Delete Employees" />

        <TextView android:id="@+id/resultText" android:textSize="16sp"
android:paddingTop="16dp" />
    </LinearLayout>
</ScrollView>
```

DatabaseHelper.java

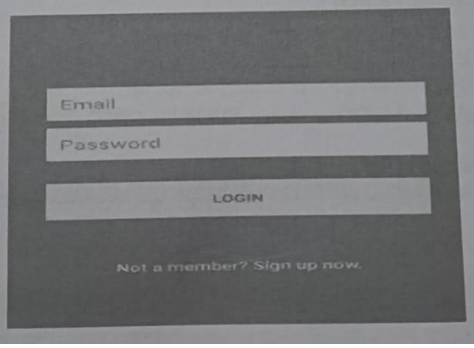
```
package com.example.empdeptapp;


import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
```

```
import android.database.sqlite.SQLiteOpenHelper;
public class DatabaseHelper extends SQLiteOpenHelper {

    public DatabaseHelper(Context context) {
        super(context, "EmpDeptDB", null, 1);
    }

    @Override
    public void onConfigure(SQLiteDatabase db) {
        super.onConfigure(db);
        db.setForeignKeyConstraintsEnabled(true);  // ✅ Important for foreign key to work
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        db.execSQL("CREATE TABLE Dept(dept_no INTEGER PRIMARY KEY, dept_name TEXT,
location TEXT)");
        db.execSQL("CREATE TABLE Emp(emp_no INTEGER PRIMARY KEY, emp_name TEXT,
address TEXT, phone TEXT, salary REAL, dept_no INTEGER, FOREIGN KEY(dept_no)
REFERENCES Dept(dept_no))");
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
        db.execSQL("DROP TABLE IF EXISTS Emp");
        db.execSQL("DROP TABLE IF EXISTS Dept");
        onCreate(db);
    }

    public void insertDept(int deptNo, String name, String location) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues cv = new ContentValues();
        cv.put("dept_no", deptNo);
        cv.put("dept_name", name);
        cv.put("location", location);
        db.insert("Dept", null, cv);
    }

    public void insertEmp(int empNo, String name, String address, String phone, double
salary, int deptNo) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues cv = new ContentValues();
        cv.put("emp_no", empNo);
        cv.put("emp_name", name);
```

```
        cv.put("address", address);
        cv.put("phone", phone);
        cv.put("salary", salary);
        cv.put("dept_no", deptNo);
        db.insert("Emp", null, cv);
    }

    public int deleteEmployeesByDeptName(String deptName) {
        SQLiteDatabase db = this.getWritableDatabase();
        Cursor cursor = db.rawQuery("SELECT dept_no FROM Dept WHERE dept_name = ?",
new String[]{deptName});
        int deleted = 0;
        if (cursor.moveToFirst()) {
            int deptNo = cursor.getInt(0);
            deleted = db.delete("Emp", "dept_no = ?", new String[]{String.valueOf(deptNo)});
        }
        cursor.close();
        return deleted;
    }
}
```

SET B

1. Create sample application with login module (Check username and password). On successful login, pass username to next screen And on failing login, alert user using Toast (Hint :Use Login(username, password) Table.)



->

<u>LoginActivity.java</u>

```
package com.example.loginmodule;

import android.content.Intent;
```

```
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class LoginActivity extends AppCompatActivity {

    EditText usernameEditText, passwordEditText;
    Button loginButton;
    DBHelper dbHelper;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_login);

        usernameEditText = findViewById(R.id.username);
        passwordEditText = findViewById(R.id.password);
        loginButton = findViewById(R.id.login_button);
        dbHelper = new DBHelper(this);

        loginButton.setOnClickListener(v -> {
            String username = usernameEditText.getText().toString().trim();
            String password = passwordEditText.getText().toString().trim();

            if (dbHelper.checkLogin(username, password)) {
                // Successful login, pass username to the next screen
                Intent intent = new Intent(LoginActivity.this, WelcomeActivity.class);
                intent.putExtra("username", username);
                startActivity(intent);
                finish();
            } else {
                // Failed login, show Toast message
                Toast.makeText(LoginActivity.this, "Invalid username or password",
Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

DBHelper

```
package com.example.loginmodule;


import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

public class DBHelper extends SQLiteOpenHelper {

   public static final String DATABASE_NAME = "userDatabase.db";
   public static final String TABLE_NAME = "Login";
   public static final String COL_1 = "username";
   public static final String COL_2 = "password";

   public DBHelper(Context context) {
      super(context, DATABASE_NAME, null, 1);
   }

   @Override
   public void onCreate(SQLiteDatabase db) {
      db.execSQL("CREATE TABLE " + TABLE_NAME + " (username TEXT PRIMARY KEY,
password TEXT)");
   }

   @Override
   public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
      db.execSQL("DROP TABLE IF EXISTS " + TABLE_NAME);
      onCreate(db);
   }

   public boolean checkLogin(String username, String password) {
      SQLiteDatabase db = this.getReadableDatabase();
      Cursor cursor = db.rawQuery("SELECT * FROM " + TABLE_NAME + " WHERE
username = ? AND password = ?",
            new String[]{username, password});
      if (cursor != null && cursor.getCount() > 0) {
         cursor.close();
         return true;
      }
      return false;
```

```
    }
}
```

MainActivity.java

```
package com.example.loginmodule;

import android.os.Bundle;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -
> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
            return insets;
        });
    }
}
```

WelcomeActivity.java

```
package com.example.loginmodule;



import android.os.Bundle;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;
```

```
public class WelcomeActivity extends AppCompatActivity {

    TextView welcomeTextView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_welcome);

        welcomeTextView = findViewById(R.id.welcome_text);

        // Get username passed from LoginActivity
        String username = getIntent().getStringExtra("username");

        if (username != null) {
            welcomeTextView.setText("Welcome, " + username);
        }
    }
}
```

activity_login.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <EditText
        android:id="@+id/username"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Username"
        android:inputType="text" />

    <EditText
        android:id="@+id/password"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Password"
        android:inputType="textPassword" />
```

```
    <Button
        android:id="@+id/login_button"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Login" />
</LinearLayout>
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World!"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

activity_welcome.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <TextView
        android:id="@+id/welcome_text"
```

```
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Welcome"
    android:textSize="20sp" />
</LinearLayout>
```

---

2. Create Table project (pno, p_name, ptype, duration) and employee (id, e_name, qulification, joindate)

Project - employee have many to many relationship.

Using database perform following operation.

1) Add new record into table.

2) Accept a project name from user and display information of employees working on the project.

->

<u>MainActivity.java</u>

```
package com.example.projectapp;

import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.ListView;
import android.widget.ArrayAdapter;
import androidx.appcompat.app.AppCompatActivity;
import java.util.ArrayList;

public class MainActivity extends AppCompatActivity {

  private DatabaseHelper dbHelper;
  private EditText editTextProjectName;
  private Button buttonAddProject, buttonAddEmployee, buttonShowEmployees;
  private ListView listViewEmployees;
```

```
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        dbHelper = new DatabaseHelper(this);

        editTextProjectName = findViewById(R.id.editTextProjectName);
        buttonAddProject = findViewById(R.id.buttonAddProject);
        buttonAddEmployee = findViewById(R.id.buttonAddEmployee);
        buttonShowEmployees = findViewById(R.id.buttonShowEmployees);
        listViewEmployees = findViewById(R.id.listViewEmployees);

        // Add Project
        buttonAddProject.setOnClickListener(view -> {
            addProject(1, "Project Alpha", "Development", 12);
            addProject(2, "Project Delta", "Research", 6);
        });

        // Add Employee
        buttonAddEmployee.setOnClickListener(view -> {
            addEmployee(101, "Anju", "MSc in Computer Science", "2023-01-01");
            addEmployee(102, "Aniket", "BSc in IT", "2022-11-15");
        });

        // Assign Employees to Project
        buttonShowEmployees.setOnClickListener(view -> {
            String projectName = editTextProjectName.getText().toString();
            ArrayList<Employee> employees = getEmployeesByProjectName(projectName);
            ArrayAdapter<Employee> adapter = new ArrayAdapter<>(this,
                    android.R.layout.simple_list_item_1, employees);
            listViewEmployees.setAdapter(adapter);
        });
    }

    // Method to add project
    private void addProject(int pno, String p_name, String ptype, int duration) {
        SQLiteDatabase db = dbHelper.getWritableDatabase();
        db.execSQL("INSERT INTO project (pno, p_name, ptype, duration) VALUES (?, ?, ?,
?)",
                new Object[]{pno, p_name, ptype, duration});
    }

    // Method to add employee
```

```
    private void addEmployee(int id, String name, String qualification, String joinDate) {
        SQLiteDatabase db = dbHelper.getWritableDatabase();
        db.execSQL("INSERT INTO employee (id, e_name, qualification, joindate) VALUES (?,
?, ?, ?)",
                new Object[]{id, name, qualification, joinDate});
    }

    // Method to assign employee to project
    private void assignEmployeeToProject(int pno, int id) {
        SQLiteDatabase db = dbHelper.getWritableDatabase();
        db.execSQL("INSERT INTO project_employee (pno, id) VALUES (?, ?)", new
Object[]{pno, id});
    }

    // Method to get employees by project name
    private ArrayList<Employee> getEmployeesByProjectName(String projectName) {
        SQLiteDatabase db = dbHelper.getReadableDatabase();
        ArrayList<Employee> employees = new ArrayList<>();

        String query = "SELECT e.id, e.e_name, e.qualification, e.joindate " +
                "FROM employee e " +
                "JOIN project_employee pe ON e.id = pe.id " +
                "JOIN project p ON pe.pno = p.pno " +
                "WHERE p.p_name = ?";
        Cursor cursor = db.rawQuery(query, new String[]{projectName});

        if (cursor.moveToFirst()) {
            do {
                int id = cursor.getInt(cursor.getColumnIndex("id"));
                String name = cursor.getString(cursor.getColumnIndex("e_name"));
                String qualification = cursor.getString(cursor.getColumnIndex("qualification"));
                String joinDate = cursor.getString(cursor.getColumnIndex("joindate"));
                employees.add(new Employee(id, name, qualification, joinDate));
            } while (cursor.moveToNext());
        }
        cursor.close();
        return employees;
    }
}
```

DatabaseHelper.java

```
package com.example.projectapp;


import android.content.Context;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

public class DatabaseHelper extends SQLiteOpenHelper {

    public static final String DATABASE_NAME = "project_employee.db";
    public static final int DATABASE_VERSION = 1;

    // SQL queries to create tables
    public static final String CREATE_PROJECT_TABLE = "CREATE TABLE project (" +
            "pno INTEGER PRIMARY KEY, " +
            "p_name TEXT, " +
            "ptype TEXT, " +
            "duration INTEGER);";

    public static final String CREATE_EMPLOYEE_TABLE = "CREATE TABLE employee (" +
            "id INTEGER PRIMARY KEY, " +
            "e_name TEXT, " +
            "qualification TEXT, " +
            "joindate TEXT);";

    public static final String CREATE_PROJECT_EMPLOYEE_TABLE = "CREATE TABLE
project_employee (" +
            "pno INTEGER, " +
            "id INTEGER, " +
            "PRIMARY KEY (pno, id), " +
            "FOREIGN KEY (pno) REFERENCES project(pno), " +
            "FOREIGN KEY (id) REFERENCES employee(id));";

    public DatabaseHelper(Context context) {
        super(context, DATABASE_NAME, null, DATABASE_VERSION);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        db.execSQL(CREATE_PROJECT_TABLE);
        db.execSQL(CREATE_EMPLOYEE_TABLE);
        db.execSQL(CREATE_PROJECT_EMPLOYEE_TABLE);
    }
```

```
    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
        db.execSQL("DROP TABLE IF EXISTS project");
        db.execSQL("DROP TABLE IF EXISTS employee");
        db.execSQL("DROP TABLE IF EXISTS project_employee");
        onCreate(db);
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <EditText
        android:id="@+id/editTextProjectName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Project Name" />

    <Button
        android:id="@+id/buttonAddProject"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Add Project" />

    <Button
        android:id="@+id/buttonAddEmployee"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Add Employee" />

    <Button
        android:id="@+id/buttonShowEmployees"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Show Employees" />

    <ListView
```

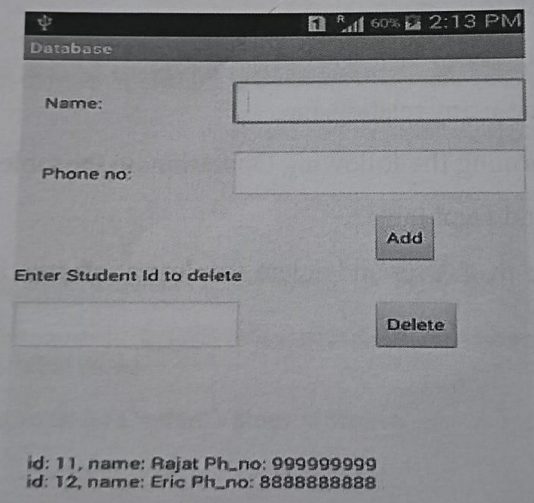
```
        android:id="@+id/listViewEmployees"
        android:layout_width="match_parent"
        android:layout_height="wrap_content" />
</LinearLayout>
```

Employee

```
package com.example.projectapp;


public class Employee {
    private int id;
    private String name;
    private String qualification;
    private String joinDate;

    public Employee(int id, String name, String qualification, String joinDate) {
        this.id = id;
        this.name = name;
        this.qualification = qualification;
        this.joinDate = joinDate;
    }

    public int getId() {
        return id;
    }

    public String getName() {
        return name;
    }

    public String getQualification() {
        return qualification;
    }

    public String getJoinDate() {
        return joinDate;
    }
}
```

SET C

1) Create simple application shown below. Create table Student(Sid ,Sname ,phno). Use autoincrement for Sid and Perform following Operation.
   a. Add Student and display its information.
   b. Delete Student.



->

MainActivity.java

```
package com.example.studentapp;


import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText etName, etPhone, etDeleteId;
    Button btnAdd, btnDelete;
    TextView tvOutput;
    StudentDatabaseHelper dbHelper;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etName = findViewById(R.id.etName);
```

```
        etPhone = findViewById(R.id.etPhone);
        etDeleteId = findViewById(R.id.etDeleteId);
        btnAdd = findViewById(R.id.btnAdd);
        btnDelete = findViewById(R.id.btnDelete);
        tvOutput = findViewById(R.id.tvOutput);

        dbHelper = new StudentDatabaseHelper(this);

        btnAdd.setOnClickListener(v -> {
            String name = etName.getText().toString();
            String phone = etPhone.getText().toString();
            dbHelper.addStudent(name, phone);
            displayStudents();
        });

        btnDelete.setOnClickListener(v -> {
            int sid = Integer.parseInt(etDeleteId.getText().toString());
            dbHelper.deleteStudent(sid);
            displayStudents();
        });

        displayStudents();
    }

    private void displayStudents() {
        String data = dbHelper.getAllStudents();
        tvOutput.setText(data);
    }
}
```

StudentDatabaseHelper.java

```
package com.example.studentapp;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

public class StudentDatabaseHelper extends SQLiteOpenHelper {

    private static final String DATABASE_NAME = "studentDB";
```

```
private static final int DATABASE_VERSION = 1;
private static final String TABLE_NAME = "student";

public StudentDatabaseHelper(Context context) {
   super(context, DATABASE_NAME, null, DATABASE_VERSION);
}

@Override
public void onCreate(SQLiteDatabase db) {
   String createTable = "CREATE TABLE " + TABLE_NAME + " (" +
         "Sid INTEGER PRIMARY KEY AUTOINCREMENT, " +
         "Sname TEXT, " +
         "phno TEXT)";
   db.execSQL(createTable);
}

@Override
public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
   db.execSQL("DROP TABLE IF EXISTS " + TABLE_NAME);
   onCreate(db);
}

// Add Student
public void addStudent(String name, String phone) {
   SQLiteDatabase db = this.getWritableDatabase();
   ContentValues values = new ContentValues();
   values.put("Sname", name);
   values.put("phno", phone);
   db.insert(TABLE_NAME, null, values);
}

// Delete Student
public void deleteStudent(int sid) {
   SQLiteDatabase db = this.getWritableDatabase();
   db.delete(TABLE_NAME, "Sid=?", new String[]{String.valueOf(sid)});
}

// Get All Students
public String getAllStudents() {
   SQLiteDatabase db = this.getReadableDatabase();
   Cursor cursor = db.rawQuery("SELECT * FROM " + TABLE_NAME, null);
   StringBuilder result = new StringBuilder();

   if (cursor.moveToFirst()) {
```

```
        do {
            int id = cursor.getInt(cursor.getColumnIndexOrThrow("Sid"));
            String name = cursor.getString(cursor.getColumnIndexOrThrow("Sname"));
            String phone = cursor.getString(cursor.getColumnIndexOrThrow("phno"));
            result.append("id: ").append(id)
                    .append(", name: ").append(name)
                    .append(" Ph_no: ").append(phone).append("\n");
        } while (cursor.moveToNext());
    }

    cursor.close();
    return result.toString();
  }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:orientation="vertical" android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:padding="16dp">

  <EditText
    android:id="@+id/etName"
    android:hint="Name"
    android:layout_width="match_parent"
    android:layout_height="wrap_content" />

  <EditText
    android:id="@+id/etPhone"
    android:hint="Phone no:"
    android:layout_width="match_parent"
    android:layout_height="wrap_content" />

  <Button
    android:id="@+id/btnAdd"
    android:text="Add"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content" />

  <EditText
    android:id="@+id/etDeleteId"
```

```
        android:hint="Enter Student Id to delete"
        android:layout_width="match_parent"
        android:layout_height="wrap_content" />

    <Button
        android:id="@+id/btnDelete"
        android:text="Delete"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content" />

    <TextView
        android:id="@+id/tvOutput"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:paddingTop="10dp"
        android:textAppearance="?android:attr/textAppearanceMedium"/>
</LinearLayout>
```